EconKit Mathematical Notes

EconKit Project

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1 Purpose

This document records the mathematical formulas implemented in EconKit and explains how each formula connects to computational economics workflows.

2 AR(1) Shock Process and Discretization

EconKit models many exogenous processes as

$$z_{t+1} = \rho z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad |\rho| < 1. \tag{1}$$

The stationary standard deviation is

$$\sigma_z = \frac{\sigma_\varepsilon}{\sqrt{1-\rho^2}}. \tag{2}$$

2.1 Tauchen

Given a grid $\{z_j\}_{j=1}^N$, transition probabilities are

$$\Pi_{ij} = \Pr\left(z_{t+1} \in \left[z_j - \frac{\Delta}{2}, z_j + \frac{\Delta}{2}\right] \mid z_t = z_i\right), \tag{3}$$

with endpoint bins handled by one-sided tails and interior bins by CDF differences. This converts a continuous shock to a finite Markov chain for Bellman iteration.

2.2 Rouwenhorst

For persistent processes, Rouwenhorst recursively constructs Π_N with

$$p = q = \frac{1+\rho}{2} \tag{4}$$

and midpoint row scaling at each recursion. This method preserves persistence and moments better than naive coarse grids when ρ is high.

3 Quadrature and Expectations

3.1 Gaussian-Hermite for Standard Normal

For $X \sim \mathcal{N}(0,1)$,

$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x)\phi(x)\,dx \approx \sum_{k=1}^{n} w_k f(x_k), \tag{5}$$

where EconKit uses transformed Hermite nodes and weights to match normal density.

3.2 Gauss-Laguerre for Exponential Weight

For integrals with e^{-x} weight on $[0,\infty)$,

$$\int_0^{\infty} e^{-x} g(x)\,dx \approx \sum_{k=1}^{n} w_k g(x_k). \tag{6}$$

For $X \sim \text{Exp}(1)$, this directly approximates $\mathbb{E}[g(X)]$.

3.3 Discrete Expectation

For finite support,

$$\mathbb{E}[X] = \sum_{i=1}^{N} p_i x_i, \tag{7}$$

implemented as a dot product after probability validation.

4 Interpolation in Dynamic Programming

Continuation values are evaluated off-grid via linear interpolation:

$$V(x) \approx (1-t)V(x_i) + tV(x_{i+1}), \qquad t = \frac{x - x_i}{x_{i+1} - x_i}. \tag{8}$$

This is central to VFI/EGM because policy updates typically generate non-grid next states.

5 Bounded Scalar Optimization (Implemented in Session 4)

Given objective $u(c) + \beta\mathbb{E}[V(x')]$, many subproblems reduce to

$$x^* = \arg\max_{x \in [a,b]} f(x). \tag{9}$$

EconKit implements:

- Golden-section search: robust derivative-free interval contraction.
- Brent-style search: combines parabolic interpolation with safeguarded bracketing.

These methods are used when derivatives are unavailable or expensive.

5.1 Golden-Section Interval Update

Given interior points $x_1 < x_2$ in $[a, b]$,

$$x_1 = a + (1 - \varphi^{-1})(b - a), \qquad x_2 = a + \varphi^{-1}(b - a), \qquad \varphi = \frac{1 + \sqrt{5}}{2}, \tag{10}$$

the interval is contracted by comparing $f(x_1)$ and $f(x_2)$, preserving the maximizer bracket while reusing one function evaluation per iteration.

5.2 Brent-Style Step Selection

Brent-style bounded search alternates between:

1. parabolic interpolation (fast local convergence when stable), and
2. golden-section fallback (robust global progress).

The safeguarded update keeps iterates inside $[a, b]$, making it suitable for economic policy problems with hard constraints.

6 Value Function Iteration and Bellman Residual (Implemented in Session 6)

Let T denote the Bellman operator on value functions. Standard VFI iterates

$$V^{(k+1)} = T\left(V^{(k)}\right), \qquad k = 0, 1, 2, \ldots \tag{11}$$

until a fixed-point tolerance is met.

6.1 Relaxed Update

To improve stability in some models, EconKit exposes relaxation $\omega \in (0, 1]$:

$$\widetilde{V}^{(k+1)} = \omega V^{(k+1)} + (1 - \omega) V^{(k)}. \tag{12}$$

When $\omega = 1$, this reduces to plain VFI.

6.2 Bellman Residual Diagnostics

The stopping metric is the residual between successive iterates, typically the sup norm:

$$r_k^{\infty} = \left\| \widetilde{V}^{(k+1)} - V^{(k)} \right\|_{\infty} = \max_i \left| \widetilde{V}_i^{(k+1)} - V_i^{(k)} \right|. \tag{13}$$

EconKit also exposes an L^2-style residual for diagnostics:

$$r_k^2 = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(\widetilde{V}_i^{(k+1)} - V_i^{(k)} \right)^2}. \tag{14}$$

Convergence is declared when $r_k^{\infty} < \varepsilon$, where ε is the configured tolerance.

6.3 Howard-Style Policy Evaluation Hook (Session 7)

Given a candidate policy $\pi^{(k)}$, policy evaluation applies repeated updates under fixed policy:

$$V^{(k,j+1)} = T^{\pi^{(k)}}\left(V^{(k,j)}\right), \qquad j = 0, 1, \ldots, H-1, \tag{15}$$

where H is the configured number of Howard policy-improvement steps. This can reduce outer Bellman iterations by doing extra work per iteration when policy evaluation is cheap relative to policy updates.

7 Canonical Consumption-Savings Helper (Implemented in Session 8)

For asset state a, deterministic income y, and gross return $1 + r$, the helper solves the standard one-dimensional Bellman problem on an asset grid:

$$V(a) = \max_{a' \in \mathcal{A}} \left\{ u(c) + \beta V(a') \right\}, \tag{16}$$

subject to

$$c = (1 + r)a + y - a', \qquad c > 0. \tag{17}$$

The implementation performs brute-force search over a' on a fixed grid to produce a robust baseline policy/value solution.

7.1 CRRA Utility

The utility specification is

$$u(c) = \begin{cases} \log(c), & \sigma = 1, \\ \dfrac{c^{1-\sigma} - 1}{1 - \sigma}, & \sigma \neq 1. \end{cases} \tag{18}$$

7.2 Residual-History Summary Diagnostics

Beyond per-iteration residuals r_k, Session 8 adds summary metrics over history $\{r_0, \ldots, r_{K-1}\}$:

$$\text{avg-log-drop} = \frac{\log_{10}(\max(r_0, \epsilon)) - \log_{10}(\max(r_{K-1}, \epsilon))}{K - 1}, \tag{19}$$

with machine-safe ϵ. This statistic gives an interpretable average orders-of-magnitude reduction per iteration.

8 Stochastic-Income Canonical Helper and Persistence (Implemented in Session 9)

With income shock state z and transition matrix Π, the stochastic consumption-savings Bellman problem is

$$V(a, z) = \max_{a' \in \mathcal{A}} \left\{ u(c) + \beta \sum_{z'} \Pi(z, z') V(a', z') \right\}, \tag{20}$$

with budget constraint

$$c = (1 + r)a + y(z) - a', \qquad c > 0. \tag{21}$$

EconKit's Session 9 canonical helper uses Tauchen or Rouwenhorst transitions to compute this expected continuation-value term on a product grid.

8.1 Convergence Artifact Persistence

Session 9 adds JSON-ready persistence for convergence diagnostics. Given history $\{r_k\}$, the artifact stores:

$$\texttt{artifact} = \{\texttt{history}, \texttt{summary}, \texttt{metadata}\}, \tag{22}$$

where `summary` includes count, endpoints, monotonicity flag, and average log-drop metric. This enables reproducible solver diagnostics across runs.

9 Liquid-Illiquid Prototype and Review Hooks (Implemented in Session 10)

Session 10 introduces a two-asset prototype with liquid ℓ and illiquid i holdings:

$$V(\ell, i) = \max_{\ell', i'} \left\{ u(c) + \beta V(\ell', i') \right\}, \tag{23}$$

subject to

$$c = (1 + r_\ell)\ell + (1 + r_i)i + y - \ell' - i' - \kappa|i' - i|, \qquad c > 0, \tag{24}$$

where κ is an optional illiquid-adjustment cost parameter.

9.1 Policy Monotonicity Diagnostic

Given a policy array π, monotonicity along an axis is checked via first differences $\Delta\pi$. Violations are counted where

$$\Delta\pi < -\tau, \tag{25}$$

with tolerance τ. This yields a compact monotonicity summary for review.

9.2 Iteration Hooks and Export

Solver iteration hooks receive per-iteration residual and current iterate, enabling online diagnostics. Convergence artifacts are exported with schema metadata for versioned downstream analysis.

10 Session 11: Simulation Slice and Pruned Two-Asset Search

Session 11 introduces a first simulation vertical slice for policy-driven Markov dynamics. Given policy map $a' = \pi(a, z)$ and transition matrix Π, simulated paths evolve by

$$a_{t+1} = \pi(a_t, z_t), \qquad z_{t+1} \sim \Pi(z_t, \cdot). \tag{26}$$

This yields reproducible simulated state sequences and summary moments.

10.1 Policy Search-Window Pruning

For the liquid-illiquid prototype, Session 11 adds optional local search around previous-iteration policy indices. If prior argmax indices are $(j^{\ell}_{t-1}, j^{i}_{t-1})$, candidate sets are restricted to

$$j^{\ell} \in [j^{\ell}_{t-1} - w,\, j^{\ell}_{t-1} + w], \qquad j^{i} \in [j^{i}_{t-1} - w,\, j^{i}_{t-1} + w], \tag{27}$$

with window size w. This reduces brute-force search cost while preserving the full-search path when w is unset.

10.2 Artifact Schema Validation

Session 11 formalizes schema checks for convergence artifacts by validating required fields and supported schema versions. Legacy artifacts without schema blocks are optionally accepted for backward compatibility.

11 Session 12: Estimation Stub and Optimization Parity

Session 12 introduces a first estimation-stub bridge from simulation moments to calibration objectives via weighted quadratic distance:

$$\mathcal{L}(\theta) = \sum_{m=1}^{M} w_m \left(\hat{g}_m(\theta) - g_m^{\text{target}}\right)^2, \tag{28}$$

where $\hat{g}_m$ are simulated moments, g_m^{target} are data targets, and w_m are moment weights.

11.1 Backend Optimization Parity Closure

Session 12 closes the strict-backend gap for bounded optimization API routing by adding Numba backend entries for golden-section and Brent-style methods, ensuring consistent backend selection semantics in strict mode.

12 Session 13: Pipeline Integration, Diagnostics, and JAX Pilot

Session 13 connects solve, simulation, and estimation into a compact reference workflow and adds simulation diagnostics for review:

$$\text{solve} \rightarrow \text{simulate} \rightarrow \hat{g}(\theta) \rightarrow \mathcal{L}(\theta). \tag{29}$$

Here, $\hat{g}(\theta)$ denotes simulated moments extracted from generated paths under parameters θ, and $\mathcal{L}(\theta)$ is the weighted quadratic distance objective.

12.1 Lag-1 Asset Autocorrelation Diagnostic

Given simulated asset path $\{a_t\}_{t=0}^{T-1}$, Session 13 reports

$$\rho_a(1) = \text{Corr}(a_t, a_{t+1}). \tag{30}$$

This summarizes persistence in simulated asset dynamics and helps identify whether policy-induced state dynamics are too diffusive or too sticky.

12.2 Empirical Shock Transition Frequencies

For simulated shock indices $z_t \in \{1, \ldots, N\}$, define empirical counts

$$C_{ij} = \sum_{t=0}^{T-2} \mathbf{1}\{z_t = i,\ z_{t+1} = j\}. \tag{31}$$

Session 13 reports row-normalized frequencies

$$\widehat{\Pi}_{ij} = \begin{cases} \dfrac{C_{ij}}{\sum_{k=1}^{N} C_{ik}}, & \sum_{k=1}^{N} C_{ik} > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{32}$$

which provide a simulation-based diagnostic against intended transition behavior.

12.3 Optional JAX Pilot Dispatch Entries

Session 13 introduces optional JAX dispatch registrations for interpolation and AR(1) discretization APIs. These entries are dependency-gated and preserve the dispatcher contract while enabling future JAX-native backend expansion.

13 Maintenance Rule

Update this document every session where numerical methods are added or changed. Each new method must include:

1. Mathematical statement,
2. Economic interpretation,
3. Numerical approximation details,
4. Link to implementation module.